

China Semiconductor Industry

How the industry works, where growth comes from, and what structurally limits it. A factual overview for long-term equity investors.

METHODOLOGY & SOURCE DISCIPLINE

Every figure below is attributed to a named external source (regulator filings, company results, industry associations, reputable analysis) dated in brackets. The 15th Five-Year Plan section quotes the official State Council Information Office (SCIO) briefing transcript directly. Items not stated in a source are marked (**inferred**). Where retail aggregator data is used, it is flagged for terminal reconciliation. Market-size and self-sufficiency figures vary materially by definition (chips vs. equipment; capacity vs. revenue) and by provider — ranges, not point estimates, are used where sources disagree.

1. Industry purpose & core economics

Integrated circuits (ICs) are the foundational components of all modern electronics. The Chinese industry exists for one structural reason: China is the world's largest maker and buyer of electronics but cannot meet its own chip demand domestically. China imported **US\$385bn** of ICs in 2024, up 10.4% year-on-year and larger than its crude-oil import bill of US\$325bn — chips are China's single largest import by value (General Administration of Customs, via SCMP, Jan 2025). The domestic semiconductor market was roughly **US\$186–218bn** in 2025, growing at about 7–8% a year (Expert Market Research; Mordor Intelligence, Jan 2026).

The economics are defined by extreme capital intensity and a value chain in which margin concentrates at the ends of difficulty. A single leading-edge fab can cost US\$15–20bn to build (TMT IB Guide, Mar 2026), and design plus front-end fabrication account for roughly 90% of value-add; assembly and test are lower-margin (SIA filing to USTR, Feb 2025). China's structural weakness is self-sufficiency: domestic firms supplied about **33%** of China's chip needs in 2024 (SEMI China, via Nikkei, Mar 2026), and semiconductor-*equipment* self-sufficiency was only ~13.6% in 2024, rising toward ~35% in 2025 (TrendForce, Feb 2025; CSIS, May 2026).

2. Industry structure & competitive edge

The industry runs on three business models — **fabless** (design only), **foundry** (manufacture to others' designs), and **IDM** (both) — plus **OSAT** (outsourced assembly, packaging and test) downstream and equipment, EDA software and materials upstream (MDPI value-chain study, 2024). China's named players span the chain: IDMs YMTC (NAND) and CXMT (DRAM); pure-play foundries SMIC and Hua Hong; fabless designers HiSilicon, UNISOC, Loongson, Cambricon and Zhaoxin; and OSAT leaders JCET, Huatian and Tongfu (Wikipedia, current).

SMIC is the largest mainland foundry and around the world's fifth-largest; it posted FY2025 revenue of **US\$9.33bn**, up ~16%, with gross margin of 21% (SMIC results, via Caixin / Power Semiconductors Weekly, Feb–Mar 2026). China's competitive edge is **not** technology leadership but a combination of state capital, a captive domestic buyer base, and price: SMIC reportedly cut 28nm foundry pricing from ~US\$2,500 to US\$1,500 per wafer (Semi Vision, via TechSoda, Dec 2024). China leads globally in OSAT (~38% of the market in 2020, multiple top-10 players — SIA, 2021) but is weak in EDA (~2% of the global market — Emyprean, via CSIS) and in equipment.

3. Upstream and downstream: the value chain

The chain runs left to right: **upstream** tools and inputs (EDA/IP, fabrication equipment, materials) feed **midstream** design and wafer fabrication, which feed **downstream** assembly/test and end demand. China's position is the mirror image of a leading-edge economy: it is strong at the two ends — the world's largest demand pool and a global OSAT leader — but import-dependent at the upstream chokepoints (EDA, lithography) and capped at advanced-node fabrication (SIA, 2021; CSIS, 2026). The colour coding below summarises that structural position.

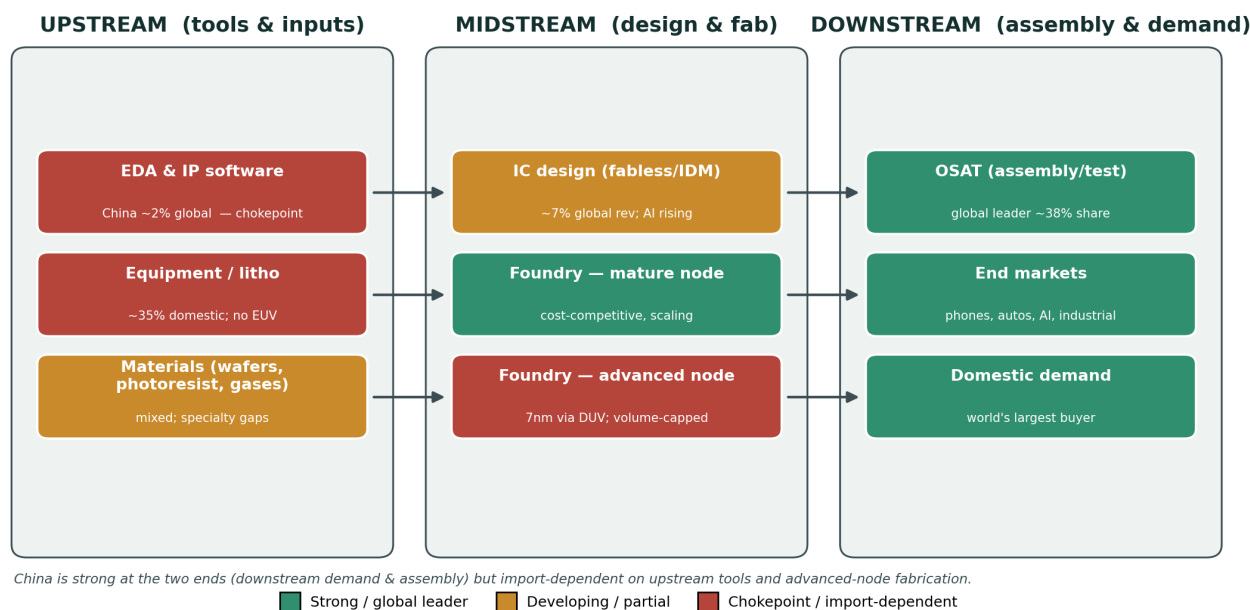


Figure 1. China's competitive position across the semiconductor value chain. Colour = strength of domestic position, not segment size. Sources: SIA (2021); CSIS (2025–26); Tianxia Gongchang (2026).

4. Demand and growth drivers

Three structural pulls drive end demand: AI adoption, data localisation and cloud build-out (DBS, Jun 2025); new-energy vehicles, where China is forecast to consume ~40% of global silicon-carbide (SiC) wafers by 2030, up from ~15% in 2023, on ~60% NEV penetration (Mordor Intelligence, Jan 2026); and 5G/RF content. But the dominant China-specific driver is **domestic substitution** — replacing the US\$385bn import bill with home-made chips. The roadmap is to lift self-sufficiency from ~20% today to **50–60% by 2030** (DBS, Jun 2025). This is policy-enforced, not purely commercial: SOE procurement guidance favours domestic chips (SASAC/NDRC, Aug 2024, via SIA), and China's MIIT reportedly told telecom operators to remove all foreign semiconductors from their networks by 2027 (CSIS, May 2026). The domestic share of China's AI-chip market is projected to reach ~50% in 2026 (TrendForce, via CSIS, 2026).

5. Supply side, cost structure and constraints

Supply economics are dominated by capital intensity and one hard constraint. SMIC's 2025 capex was **US\$8.1bn** (+10.5%), and depreciation ran at ~42% of revenue in Q3 2025 — a fixed cost absorbed before anything else (SMIC results; Kristal.ai, Nov 2025). That is why margins sit at 19–22% even at ~96% utilisation, far below leading-edge foundry economics. The binding constraint is **lithography**: ASML is the sole maker of EUV tools and exports to China are banned, while the leading domestic stepper (SMEE) covers only ~90nm DUV — roughly a generation behind 14nm (Tianxia Gongchang, 2026). DUV multi-patterning lets SMIC produce 7nm, but at volumes constrained to the “low tens of thousands” of wafers a month rather than the hundreds of thousands originally planned (CSIS, via Law & Economics Center, Mar 2025).

Dependence on foreign tools remains structurally high: China imported ~US\$38.5bn of semiconductor equipment in 2024 (+19%), with Japan and the Netherlands supplying the majority (Wells Fargo customs data, via R. Castellano, Apr 2025; gtaic, 2025). EDA is a near-US monopoly (Synopsys, Cadence). A talent gap compounds it — China needed ~199,300 IC professionals in 2022 against ~164,300 employed (Mordor Intelligence, Jan 2026).

6. Technology changes

With EUV blocked, technology effort has rerouted around the chokepoint. SMIC uses DUV multi-patterning and self-aligned quadruple patterning (SAQP) to push toward 5nm-class logic (Grokikipedia, Mar 2026). **Advanced packaging and chiplets** are emerging as a way to gain performance without a smaller node — Huawei has patented quad-chiplet GPU designs (CRS, Sep 2025). **Compound semiconductors** (SiC, GaN) are a node-agnostic arena where China is already cost-competitive: a 6-inch SiC wafer that cost ~US\$1,500 from a Western leader is sold near US\$500 by Chinese producers (Nikkei, via Tom's Hardware, Feb 2025). Domestically,

the equipment “Big Three” — NAURA, AMEC and ACM Research — hold a combined ~45–50% of the home market, with etch and deposition localisation reaching 30–45% (Tianxia Gongchang, 2026; CSIS, 2026). FY2024 revenue ran ~RMB 27.6–31.8bn at NAURA and ~RMB 9.1bn at AMEC (company guidance, via TrendForce, Jan 2025).

7. Regulatory change: the 15th Five-Year Plan (2026–2030)

The 15th Five-Year Plan elevates the sector explicitly. In the SCIO briefing transcript, NDRC official Bai Jingyu names the sector among medium-term *emerging pillar industries*:

15TH FIVE-YEAR PLAN — SCIO BRIEFING TRANSCRIPT (PRIMARY SOURCE)

[Transcript, p.13]: the plan will “focus on fostering emerging pillar industries, including integrated circuits, biomedicine, and aerospace, to form new pillars to support the national economy.”

[Transcript, p.15]: it targets “decisive breakthroughs in key and core technologies in priority fields such as integrated circuits” to enhance the “self-reliance and controllability of industrial chains.”

[Transcript, p.14]: a national “AI Plus” initiative and a “national integrated computing power network” anchor demand for advanced compute.

Read against external reporting, this confirms a continuity of priority rather than a new departure. Industry executives have proposed a 2026–2030 roadmap targeting **80% self-sufficiency by 2030**, a fully domestic-equipment 7nm line, and stable 14nm production (Nikkei, Mar 2026). The financing vehicle is the third National IC Industry Investment Fund (“Big Fund III”), established May 2024 with RMB 344bn (~US\$47.5bn) and tilted toward equipment and materials — the previously under-funded chokepoints (Tianxia Gongchang, 2026). Independent analysts expect domestic-substitution mandates to *accelerate* under the plan (SIA WTO-compliance filing, Sep 2025).

8. Structural change: export markets and the mature-node glut

The defining structural shift is a pivot from importing advanced chips to flooding the mature-node market. Blocked below 14nm, China has poured capital into 28nm-and-above capacity. Its share of global mature-node foundry capacity is on a steep climb and is projected to overtake Taiwan by 2027 (TrendForce, via Economy.ac, Jan 2026). Of 97 new fabs scheduled worldwide for 2023–2025, 57 are in China (SEMI, via Economy.ac, 2026). Meanwhile China’s IC *exports* reached US\$159bn in 2024 (+17.4%), nearly double the 2018 level (Customs, via SCMP, Jan 2025) — the legacy-chip flood is already visible in trade data.

Mature-node foundry capacity share	2024	2025E	2027E	Source
China (global capacity)	~34%	—	> Taiwan	TrendForce / Economy.ac (2026)
China (% of world mature output)	~28%	~28%	~39%	IDC, via Tom’s Hardware (2025)
Taiwan (global mature capacity)	~43%	—	overtaken	TrendForce (2024/26)
China chip self-sufficiency (all)	~33%	—	50–60% by 2030	Nikkei/SEMI; DBS (2025–26)

Table 1. Mature-node positioning. Definitions differ across providers (installed capacity vs. output share); treat as directional, reconcile point cells against Bloomberg/Wind.

Is there a release valve for the glut? The structural answer is domestic absorption: China consumes most of what it builds (SMIC and Hua Hong are now ~80% domestic-customer — Renaissance Fish, Feb 2025), so import substitution of the US\$385bn bill soaks up much of the new supply. Analysts are split on whether the build-out is true “overcapacity.” CSIS argues the label is misleading because mature node is not a single market and foundries build to contracted demand; A-share chip inventories fell from ~9.5 to ~6.7 months over 2023–24 (CSIS, 2024; Renaissance Fish, 2025). Against that, Western trade bodies warn of subsidised dumping and have triggered a US Section 301 probe (USTR, Dec 2024). **(inferred)** Foundry consolidation and China’s “anti-involution” campaign may further rationalise capacity.

9. Medium-term outlook (5–10 years)

The most defensible base case is bifurcation. In **mature nodes**, China is likely to become the dominant global supplier on cost and scale, with persistent margin pressure and recurring oversupply scares. In **advanced nodes**, China is estimated to remain ~5–7 years behind the leaders and volume-capped by the lithography gap unless a domestic EUV-class tool emerges (Grokopedia, 2026). At the device level, Yole Group projects China's domestic device industry reaching ~10% of the global market and ~US\$100bn of revenue by 2030 (Yole, Nov 2025), while the self-sufficiency roadmap targets 50–60% (DBS, 2025). The investable thesis is therefore localisation share-gain inside a protected market, not global leadership.

DATA-RECONCILIATION CAVEAT

Company financials here follow reported results; market-size, capacity-share and self-sufficiency figures are drawn from third-party analysts whose definitions differ. Before any position-sizing, reconcile the starred cells — mature-node share, self-sufficiency %, and segment market sizes — against **Bloomberg** or **Wind**, and confirm SMIC/Hua Hong figures against primary filings (HKEX / SSE STAR).

Investor synthesis

Core economic engine —

import substitution of a US\$385bn chip bill. The engine is a captive, policy-guaranteed domestic market, not global cost or technology competitiveness.

Primary growth lever —

localisation mandates (state procurement plus “remove foreign chips” directives) lifting domestic share from ~33% toward 50–60% by 2030, across chips, equipment and EDA.

Structural constraints investors underestimate —

(i) the lithography chokepoint caps advanced-node *scale* regardless of how much capital is deployed; (ii) mature-node margins are being competed away — high utilisation does not mean high profit (SMIC ran ~96% utilisation with margins falling to ~19%), and depreciation is a structural drag.

Key risk that could change trajectory —

a domestic EUV-class lithography breakthrough would re-rate advanced-node names sharply; conversely, deeper export controls (node-agnostic tools, materials, EDA) or mature-node trade barriers (Section 301 tariffs) could close the export valve and expose the oversupply.

What kind of companies tend to win —

firms riding the localisation wave with pricing *protected* by either a technology moat at a chokepoint (domestic equipment, EDA, AI-accelerator leaders) or guaranteed captive demand — rather than commodity mature-node foundries fully exposed to the price war. (**inferred**) from the structural analysis above; not a recommendation on any specific security.

Selected sources

- General Administration of Customs, via South China Morning Post, “China's 2024 chip imports surged 10.4% to US\$385bn” (Jan 2025).
- SCIO / NDRC briefing transcript on the draft Outline of the 15th Five-Year Plan (2026–2030) — project document, pp. 13–15.
- DBS, “China Semiconductor: crisis breeds opportunity” (Jun 2025).
- Mordor Intelligence, China Semiconductor Device Market (Jan 2026); Expert Market Research, China Semiconductor Market (2025).
- SMIC FY2025 results, via Caixin Global, Power Semiconductors Weekly, Futunn, kr-asia (Feb–Mar 2026); Kristal.ai (Nov 2025).
- Semiconductor Industry Association, “Taking Stock of China's Semiconductor Industry” (2021); SIA WTO-compliance filing to USTR (Sep 2025).
- CSIS, “China's Localization Drive in Semiconductors” (May 2026); “Legacy Chip Overcapacity: Myth and Reality” (2024).
- Congressional Research Service R48642, “U.S. Export Controls and China: Advanced Semiconductors” (Sep 2025).
- TrendForce (Feb 2025; Mar 2026); Nikkei, via Tom's Hardware / TrendForce on mature-node share and self-sufficiency.
- Tianxia Gongchang Research, China Semiconductor Equipment Market 2026; company guidance for NAURA / AMEC / ACM (via TrendForce, Jan 2025).
- Yole Group, China Semiconductor Industry 2025 (Nov 2025); Wells Fargo customs data via R. Castellano (Apr 2025); gtaic (2025).